Systems of Linear Equations (AA HL)

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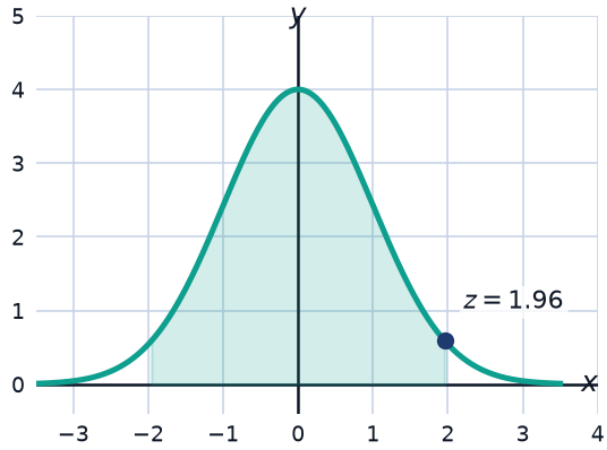
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Augmented Matrices and Gaussian Elimination

Solve the first-order linear differential equation $\frac{dy}{dx} + \frac{2}{x}y = 4x$ for $x > 0$, given the initial condition $y(1) = 2$.

2 [Foundational] • Target: Geometric Interpretation of 3 x 3 Systems

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.



3 [Foundational] • Target: Determinants and Cramer's Rule

Solve the second-order homogeneous linear differential equation $y'' - 5y' + 6y = 0$ satisfying initial conditions $y(0) = 1$ and $y'(0) = 3$.

4 [Foundational] • Target: Parameterized Solutions and Free Variables

Consider the points $A(1, -2, 3)$, $B(4, 0, 5)$, and $C(p, 1, 2)$. If the vectors $\vec{AB}$ and $\vec{AC}$ are perpendicular, find the value of p.

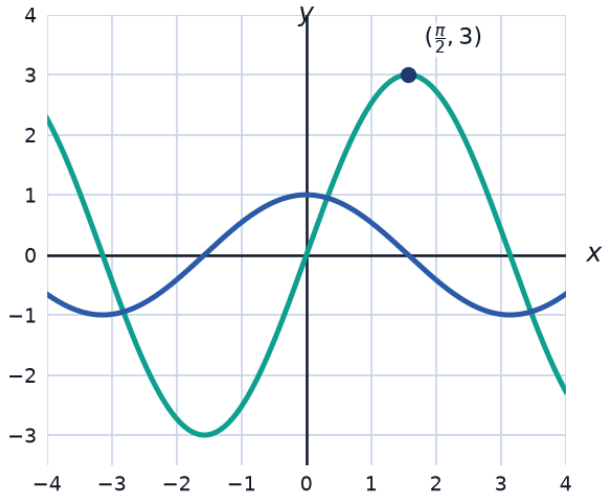
5 [Foundational] • Target: Augmented Matrices and Gaussian Elimination

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

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6 [Foundational] • Target: Geometric Interpretation of 3 x 3 Systems

Phone calls arrive at an emergency center according to a Poisson process with an average rate of 3 calls per minute. Find: (a) the probability that at least 2 calls arrive in a 1-minute period; (b) the probability that exactly 5 calls arrive in a 2-minute period.



7 [Foundational] • Target: Determinants and Cramer's Rule

A discrete random variable X takes values $0, 3, 4$ with probabilities $\frac{1}{10}, \frac{3}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Parameterized Solutions and Free Variables

The region bounded by $y = \ln x$, the line $y = 0$, the line $y = 1$, and the y-axis is rotated 2π radians about the y-axis. Find the exact volume of the resulting solid.

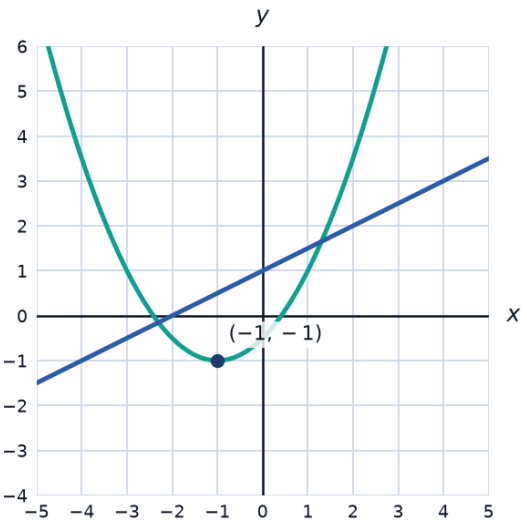
9 [Intermediate] • Target: Augmented Matrices and Gaussian Elimination

Consider the function $f(x) = \frac{1}{2}x^2 - 3x + 5$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.

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10 [Intermediate] • Target: Geometric Interpretation of 3 x 3 Systems

Find $\int xe^{2x}\,dx$ (in terms of x, with constant C).



11 [Intermediate] • Target: Determinants and Cramer's Rule

Find the number of solutions of $\sin(2\theta) = \frac{1}{2}$ in the interval $0^\circ \le \theta < 360^\circ$.

12 [Intermediate] • Target: Parameterized Solutions and Free Variables

How many solutions does $\sin\theta = \sin 30^\circ$ have in the interval $0^\circ \le \theta < 360^\circ$? Show your complete algebraic work.

13 [Intermediate] • Target: Augmented Matrices and Gaussian Elimination

How many solutions does $\sin\theta = \sin 60^\circ$ have in the interval $0^\circ \le \theta < 360^\circ$? Show your complete algebraic work.

14 [Intermediate] • Target: Geometric Interpretation of 3 x 3 Systems

How many solutions does $\sin\theta = \sin 45^\circ$ have in the interval $0^\circ \le \theta < 360^\circ$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

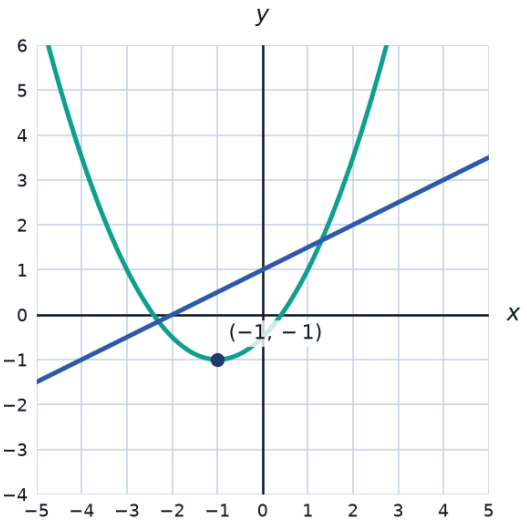
15 [Challenging] • Target: Determinants and Cramer's Rule • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

For which value of m does the system $x + my = 3,\ 2x + 4y = 6$ have infinitely many solutions? Show your complete algebraic work.

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16 [Challenging] • Target: Parameterized Solutions and Free Variables • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

How many solutions does the system $x + y = 4,\ 2x + 2y = 8$ have? Show your complete algebraic work.



17 [Challenging] • Target: Augmented Matrices and Gaussian Elimination • Cross-Lesson: 1.04 Partial Fractions

An investment grows by $3\%, 1\%, -5\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

18 [Challenging] • Target: Geometric Interpretation of 3 x 3 Systems • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

An investment grows by $-6\%, -10\%, 4\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

19 [Challenging] • Target: Determinants and Cramer's Rule • Cross-Lesson: 3.01 3D Vectors & the Cross Product

An investment grows by $-3\%, 2\%, -8\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

20 [Challenging] • Target: Parameterized Solutions and Free Variables • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

An investment grows by $1\%, -6\%, 1\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Augmented Matrices and Gaussian Elimination

In row-reducing an augmented matrix, which row operation is NOT permitted?

a. Squaring all elements in a row

b. Swapping two rows

c. Multiplying a row by a non-zero scalar

d. Adding a multiple of one row to another row

22 [Foundational] • Target: Augmented Matrices and Gaussian Elimination

An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 5]$. What does this indicate about the system?

a. The system is inconsistent (has no solutions)

b. The system has infinitely many solutions

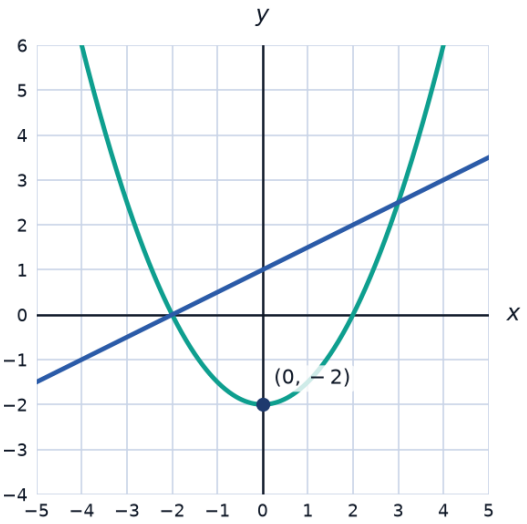
c. The system has a unique solution with $z = 5$

d. The system has trivial solutions only

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23 [Foundational] • Target: Augmented Matrices and Gaussian Elimination

An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 0]$ with 2 leading pivots across 3 variables. What does this signify?



a. The system has infinitely many solutions parameterized by 1 free variable

b. The system has a unique solution

c. The system has no solutions

d. The determinant of the matrix is 1

24 [Foundational] • Target: Geometric Interpretation of 3 x 3 Systems

What is the geometric configuration of three linear equations in three variables that possess a unique solution?

a. Three planes intersecting at a single unique point

b. Three planes intersecting along a common line

c. Three parallel non-intersecting planes

d. Three planes forming a triangular prism

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25 [Foundational] • Target: Geometric Interpretation of 3 x 3 Systems

If three planes intersect along a single straight line, what type of system is this?

a. Consistent dependent system with infinitely many solutions

b. Inconsistent system with no solutions

c. Consistent independent system with 1 solution

d. Homogeneous system with trivial solution only

26 [Foundational] • Target: Geometric Interpretation of 3 x 3 Systems

Three planes form the faces of an open triangular prism with no common line or point. How is this system classified?

a. Inconsistent (no solutions)

b. Infinitely many solutions

c. Unique solution

d. Singular dependent

27 [Foundational] • Target: Determinants and Cramer's Rule

Evaluate the 2×2 determinant $\det[[4, -3], [2, 5]]$.

a. 26

b. 14

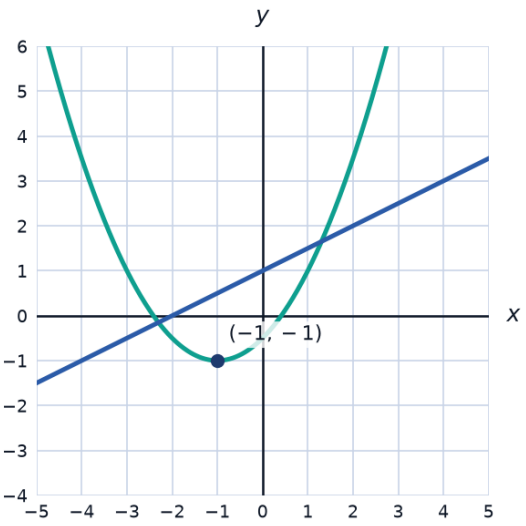
c. -26

d. 20

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28 [Foundational] • Target: Determinants and Cramer's Rule

Under what condition does a 3×3 system $A\mathbf{x} = \mathbf{b}$ have a unique solution?



a. $\det(A) \neq 0$

b. $\det(A) = 0$

c. $\mathbf{b} = \mathbf{0}$

d. $\text{Tr}(A) > 0$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Determinants and Cramer's Rule

If $\det(A) = 0$ for a system $A\mathbf{x} = \mathbf{b}$, what can be concluded?

a. The system either has no solutions or infinitely many solutions

b. The system must have no solutions

c. The system must have infinitely many solutions

d. The system has exactly 2 solutions

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30 [Intermediate] • Target: Parameterized Solutions and Free Variables

Solve the system: $x + y + z = 6$ and $2x - y + z = 3$. Parameterize the solution line using $z = \lambda$.

a. $x = 3 - \frac{2}{3}\lambda, y = 3 - \frac{1}{3}\lambda, z = \lambda$

b. $x = 3 + \lambda, y = 3 - \lambda, z = \lambda$

c. $x = 9 - 2\lambda, y = \lambda, z = \lambda$

d. $x = 1, y = 2, z = 3$

31 [Intermediate] • Target: Parameterized Solutions and Free Variables

For what value of k does the system $x + 2y = 4$ and $3x + 6y = k$ have infinitely many solutions?

a. $k = 12$

b. $k = 4$

c. $k = 0$

d. $k = 6$

32 [Intermediate] • Target: Parameterized Solutions and Free Variables

If the homogeneous system $A\mathbf{x} = \mathbf{0}$ has non-trivial solutions (solutions other than $\mathbf{x} = \mathbf{0}$), what must be true about $\det(A)$?

a. $\det(A) = 0$

b. $\det(A) > 0$

c. $\det(A) = 1$

d. $\det(A) < 0$

33 [Intermediate] • Target: Augmented Matrices and Gaussian Elimination

An investment grows by 6%, -4%, -10% in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. -4.89\%

b. -0.89\%

c. -2.89\%

d. -1.39\%

34 [Intermediate] • Target: Geometric Interpretation of 3 x 3 Systems

An investment grows by 3%, 2%, 18% in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 8.93\%

b. 5.93\%

c. 9.43\%

d. 7.43\%

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35 [Intermediate] • Target: Determinants and Cramer's Rule

An investment grows by $16\%, 4\%, -5\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 7.65\%

b. 1.65\%

c. 4.65\%

d. 3.15\%

36 [Intermediate] • Target: Parameterized Solutions and Free Variables

An investment grows by $13\%, -7\%, 6\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 1.66\%

b. 6.66\%

c. 2.16\%

d. 3.66\%

37 [Intermediate] • Target: Augmented Matrices and Gaussian Elimination

An investment grows by $1\%, 12\%, 0\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 4.19\%

b. 7.19\%

c. 5.69\%

d. 1.19\%

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric Interpretation of 3 x 3 Systems • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

An investment grows by $6\%, -1\%, 8\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 2.76\%

b. 4.26\%

c. 1.26\%

d. 2.26\%

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39 [Challenging] • Target: Determinants and Cramer's Rule • Cross-Lesson: 3.01 3D Vectors & the Cross Product

An investment grows by $14\%, 7\%, -3\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 2.77\%

b. 5.77\%

c. 8.77\%

d. 4.27\%

40 [Challenging] • Target: Parameterized Solutions and Free Variables • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

An investment grows by $10\%, 5\%, 5\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 5.14\%

b. 8.14\%

c. 6.64\%

d. 9.64\%

41 [Challenging] • Target: Augmented Matrices and Gaussian Elimination • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

An investment grows by $0\%, 19\%, -3\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 6.40\%

b. 4.90\%

c. 6.90\%

d. 7.90\%

42 [Challenging] • Target: Geometric Interpretation of 3 x 3 Systems • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Which value of k makes the system $x + 2y = 5,\ 2x + 4y = k$ have infinitely many solutions?

a. 10

b. 5

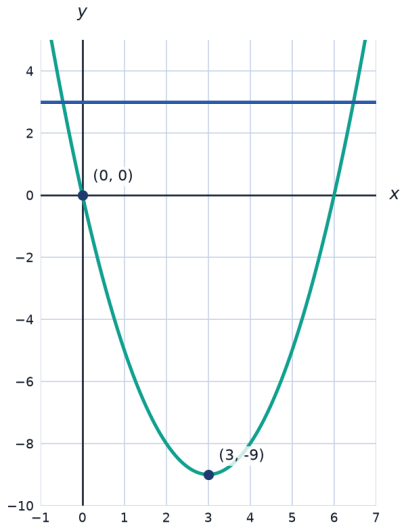
c. 20

d. 15

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43 [Challenging] • Target: Determinants and Cramer's Rule • Cross-Lesson: 4.02 The Poisson Distribution

The graph of the function $f(x) = ax^2 - 6x + c$ is shown below. The graph has a vertex at the point (3, -9) and passes through the point (0, 0). Determine the values of a and c.



a. $a = 1, c = 0$

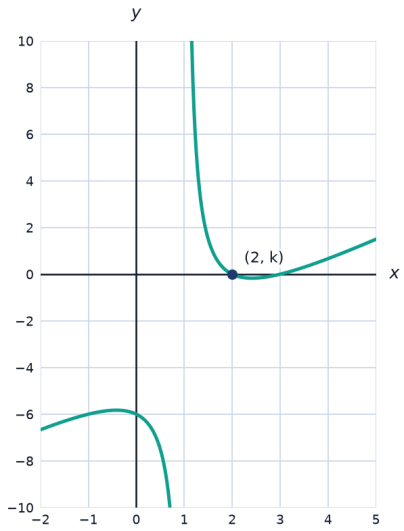
b. $a = 2, c = 0$

c. $a = 1, c = 3$

d. $a = 2, c = 3$

44 [Challenging] • Target: Parameterized Solutions and Free Variables • Cross-Lesson: 5.01 Further Integration Techniques

The graph of the function $f(x) = \frac{x^2 - ax + 6}{x - 1}$ is shown below. The graph has a vertical asymptote at $x = 1$ and a point of discontinuity at $(2, k)$. Find the value of a.



a. 4

b. 5

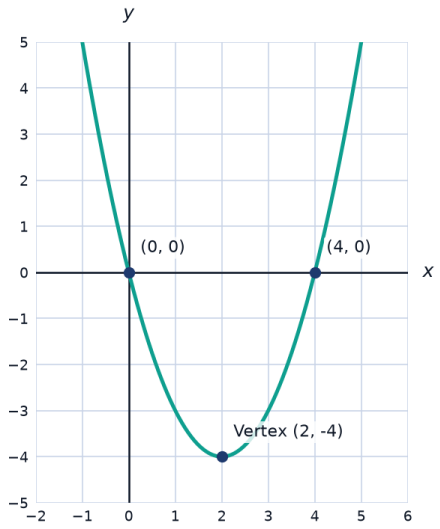
c. 6

d. 7

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45 [Challenging] • Target: Augmented Matrices and Gaussian Elimination • Cross-Lesson: 5.02 Differential Equations

The graph of the function $y = f(x)$ is shown below. The function is of the form $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(2, -4)$. The parabola passes through the points $(0, 0)$ and $(4, 0)$. Determine the value of a.



a. -1

b. 1

c. -2

d. 2

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